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Live Digital Forensics
&
AI Assisted Investigation

Introduction to Live Digital Forensics

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Live Digital Forensics is the process of acquiring and analyzing digital evidence from a computer system while it is still running. Unlike traditional forensic imaging, which focuses on powered-off devices, live forensics captures volatile and transient data that would otherwise be lost when the system is shut down.

This process involves the careful collection of critical information such as:

- ➔ Running processes and active user sessions
- ➔ RAM (volatile memory) contents
- ➔ Network connections and communication artifacts
- ➔ System logs and temporary files
- ➔ Active storage and system state information

The primary objective of live evidence acquisition is not merely to access data, but to do so in a manner that:

- ★ Preserves the integrity and current state of the live system.
- ★ Enables comprehensive forensic analysis without altering or contaminating evidence.
- ★ Maintains a documented chain of custody to ensure the evidence remains admissible and defensible in legal and investigative proceedings.

Dead vs Live Forensics

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Characteristic	Live Acquisition	Dead Acquisition
System state	Powered on	Powered off
Type of data collected	Volatile (RAM, processes, connections)	Persistent (disk, files, logs)
Risk of alteration	High (requires active intervention)	Low (with proper tools)
Legal value	Admissible if well documented	More robust for legal proceedings
Required expertise	High	Medium

Importance of Live forensics

- ➔ **Detecting fileless and memory-resident malware**
- ➔ **Capturing active network connections**
- ➔ **Identifying active user sessions and attacker presence in real time.**
- ➔ **Recovering encryption keys, credentials, and authentication tokens**
- ➔ **Understanding the current state of compromise**
- ➔ **Supporting rapid incident response**,

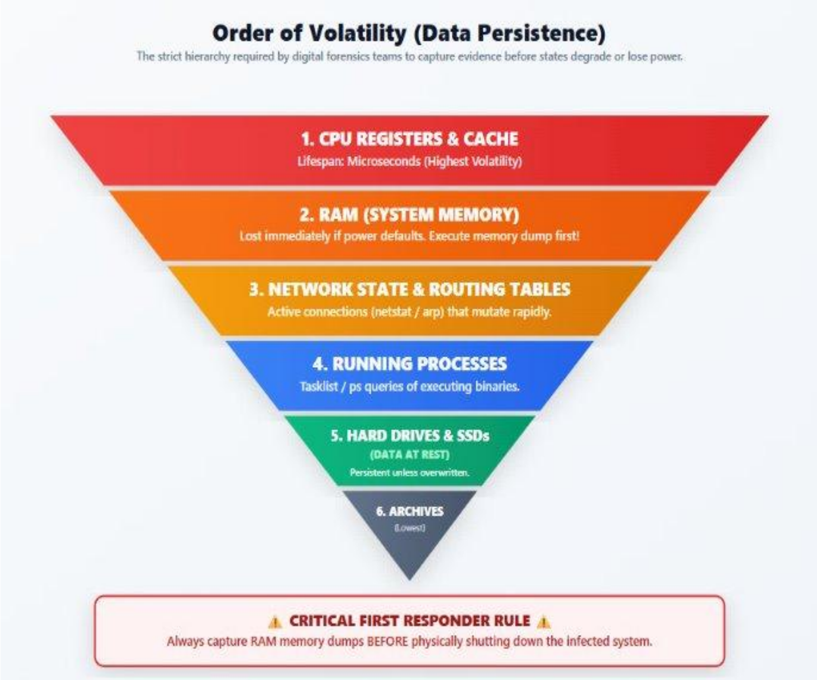
Order of Volatility

The collection of evidence should start with the most volatile item and end with least volatile item.

According to the IETF, the Order of Volatility is as follows:

1. Registers, Cache
2. Routing Table, ARP Cache, Process Table, Kernel Statistics, Memory
3. Temporary File Systems
4. Disk
5. Remote Logging and Monitoring Data
6. Physical Configuration, Network Topology
7. Archival Media

This guides investigators on what evidence to collect first. Data at the top can disappear within seconds or even nanoseconds, while data at the bottom may remain available for months or years.



Memory Forensics - Memory structure

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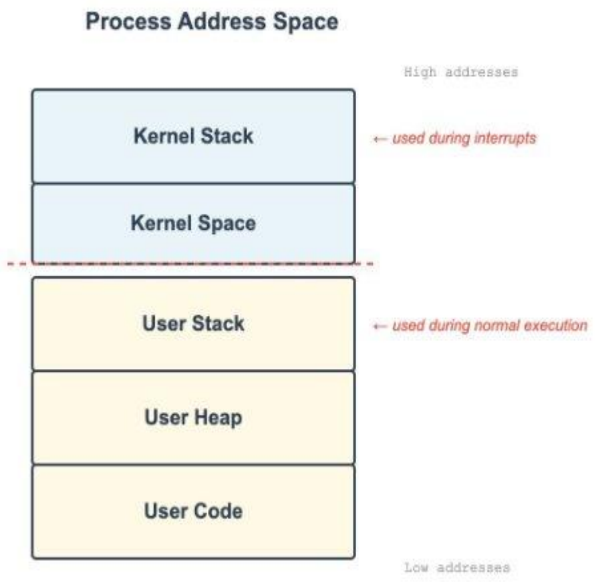
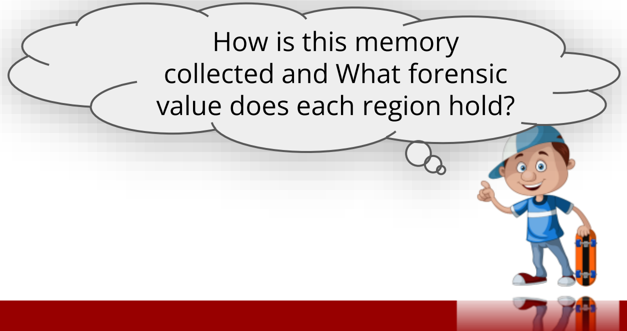


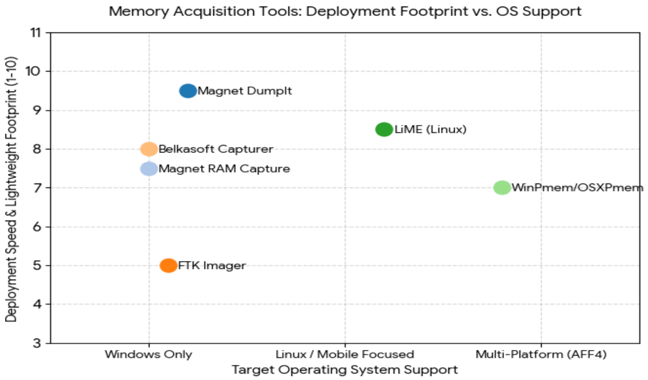
- ➔ Since RAM is among the most valuable and volatile sources of evidence, it becomes a primary focus during live forensic investigations.
- ➔ Understanding how memory is organized helps investigators identify anomalies, injected code, and suspicious behavior.
- ➔ System memory is divided into different regions used by the operating system and applications.
  - Kernel Space - Operating system code and drivers
  - User Space - Running applications and user data
  - Stack - Function calls and temporary variables
  - Heap - Dynamically allocated application memory
  - Process Memory - Executable code and process data



Memory Acquisition

Memory acquisition is the process of creating a copy of system RAM for analysis. This can be performed:

- locally on the machine
- remotely across the network
- using software tools
- using dedicated hardware solutions

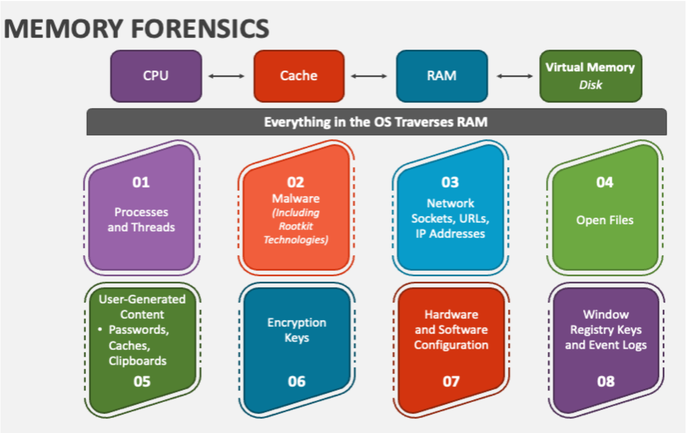


Tool for acquisition	Speed	System Impact	Best For
DumpIt	Very Fast	Low	Quick RAM capture
FTK Imager	Moderate	Medium	Beginner investigations
Magnet RAM Capture	Fast	Low	Incident response
LiME	Moderate	Low	Linux memory capture

Memory Analysis

Memory analysis is the process of extracting and interpreting forensic evidence from a captured memory dump.

Tool for analysis	Speed	System Impact
Volatility	Moderate	Deep forensic analysis
Rekall	Fast	Incident response
Autopsy	Moderate	GUI-based analysis



Memory analysis tools provide powerful forensic capabilities, but investigators often face several challenges:

- Manual memory acquisition from endpoints
- Collecting multiple artifacts from different systems
- Time-consuming investigation workflows
- Limited scalability in large environments
- Need for centralized evidence collection

What if evidence collection and analysis could be automated?

Introducing Velociraptor

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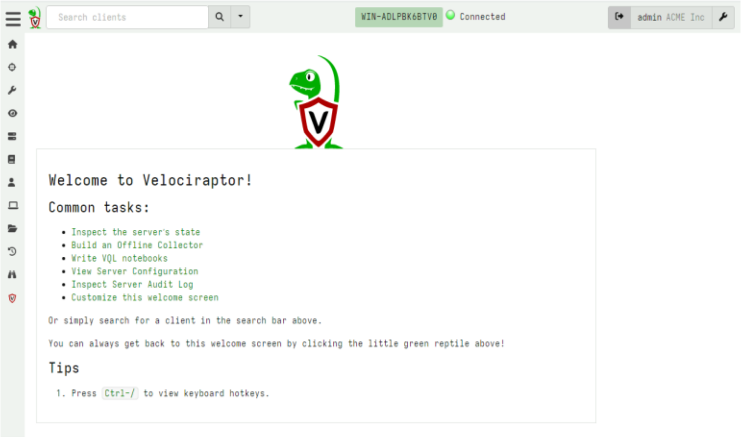
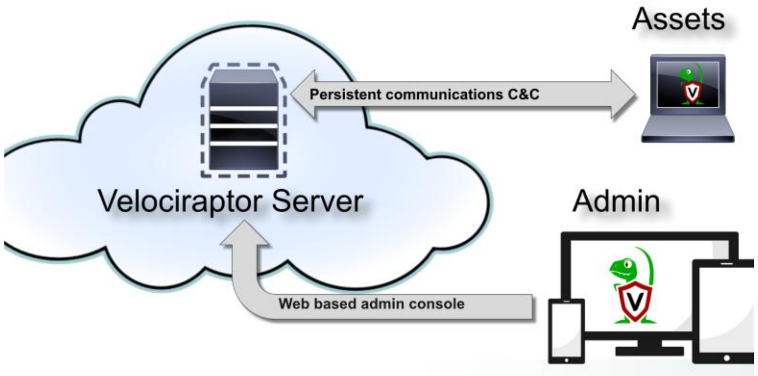
Velociraptor is an enterprise-grade, open-source DFIR platform designed for endpoint monitoring, digital forensics, and incident response.

Why Velociraptor?

- ➢ Supports Windows, Linux, macOS, and FreeBSD
- ➢ Covers the entire attack lifecycle
    - ○ Threat Hunting
    - ○ Incident Response
    - ○ Digital Forensics
    - ○ Continuous Monitoring
- ➢ Built for enterprise-scale investigations
    - ○ Single server can manage 10,000–15,000 endpoints
    - ○ Optimized for speed and scalability
    - ○ Concurrency controls ensure stability
- ➢ Uses Velociraptor Query Language (VQL)
    - ○ Create custom forensic artifacts
    - ○ Detect emerging threats rapidly
    - ○ Automate evidence collection
- ➢ Hunt at Scale
    - ○ Collect artifacts from thousands of endpoints
    - ○ Execute investigations within minutes





Key capabilities

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Collection

Remote evidence collection
Supports offline collectors
Integrates with existing EDR

Analysis

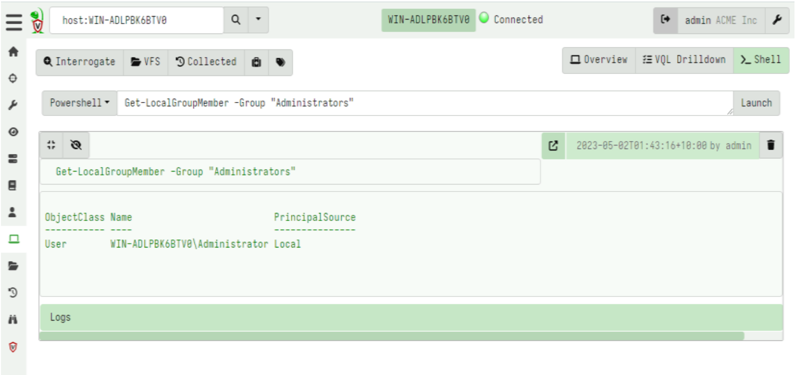
Interactive notebook for collaborative investigations
Supports in-depth forensic analysis and threat hunting
Supports external platform analysis such as timesketch

Exports & Integrations

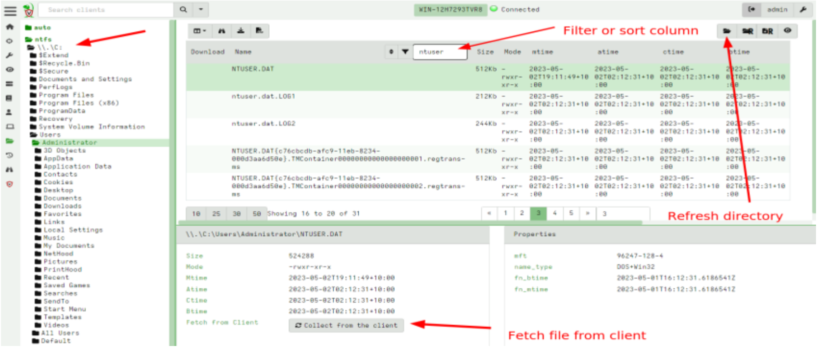
Export collected evidence as ZIP archives
Cloud backup support
Real-time integration with SIEM and data lake platforms like Splunk and elastic

Automation

Can be fully automated using a gRPC based API.
Facilitates SOAR integration and interoperability with security tools



Run shell commands and fetch files interactively



VQL query

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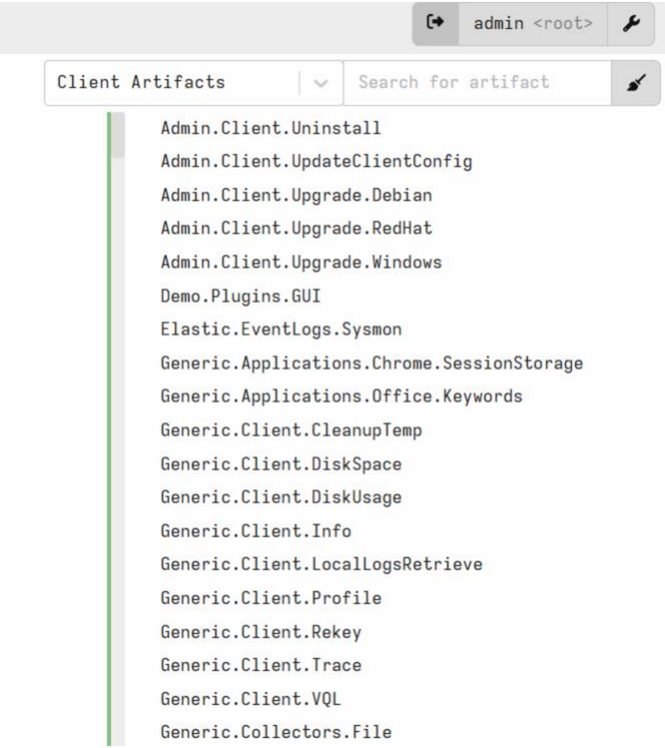
Velociraptor Query Language (VQL) is the query language used by Velociraptor to collect, filter, process, and analyze forensic artifacts from endpoints.

- It acts as the engine behind Velociraptor's artifact collection, threat hunting, and incident response capabilities
- Able to dynamically adapt to changing requirements - without needing to rebuild clients or servers.
- Enables rapid detection of new threats and Indicators of Compromise (IoCs)
- Allows investigators to create custom forensic artifacts without modifying source code
- Supports large-scale threat hunting across thousands of endpoints
- Facilitates sharing of detection logic and forensic knowledge within the DFIR community

Artifacts



- ➔ Artifacts are VQL modules
- ➔ VQL is very powerful but it is hard to remember and type a query each time.
- ➔ An Artifact is a way to document and reuse VQL queries
- ➔ Artifacts are geared towards collection of a single type of information
- ➔ Artifacts accept parameters with default values so they can be customized on each execution without needing to change any code.

Velociraptor Artifacts

Velociraptor comes with a large number of artifact types

a. Client Event artifacts monitor the endpoint
b. Server Artifacts run Client Artifacts run on the endpoint
c. on the server
d. Server Event artifacts monitor for events on the server.

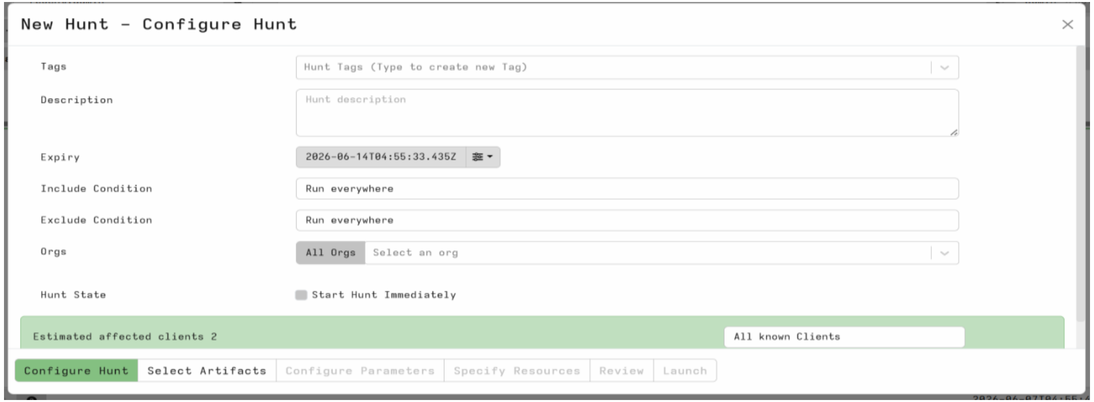
Examples of artifacts

- ❖ Windows.EventLogs.Evtx
- ❖ Windows.Forensics.Prefetch
- ❖ Windows.NTFS.MFT
- ❖ Windows.Sys.Users

Threat hunting

Threat Hunting - is the proactive process of searching for signs of malicious activity within systems and networks before alerts or security tools detect them.

1. Identify potential threat vectors.
2. Choose the correct VQL artifact to gather data.
3. Run queries across thousands of machines simultaneously.
4. Review results via GUI for malicious Indicators of Compromise (IOCs).
5. Use Velociraptor to kill processes or delete files



AI Assisted Forensics

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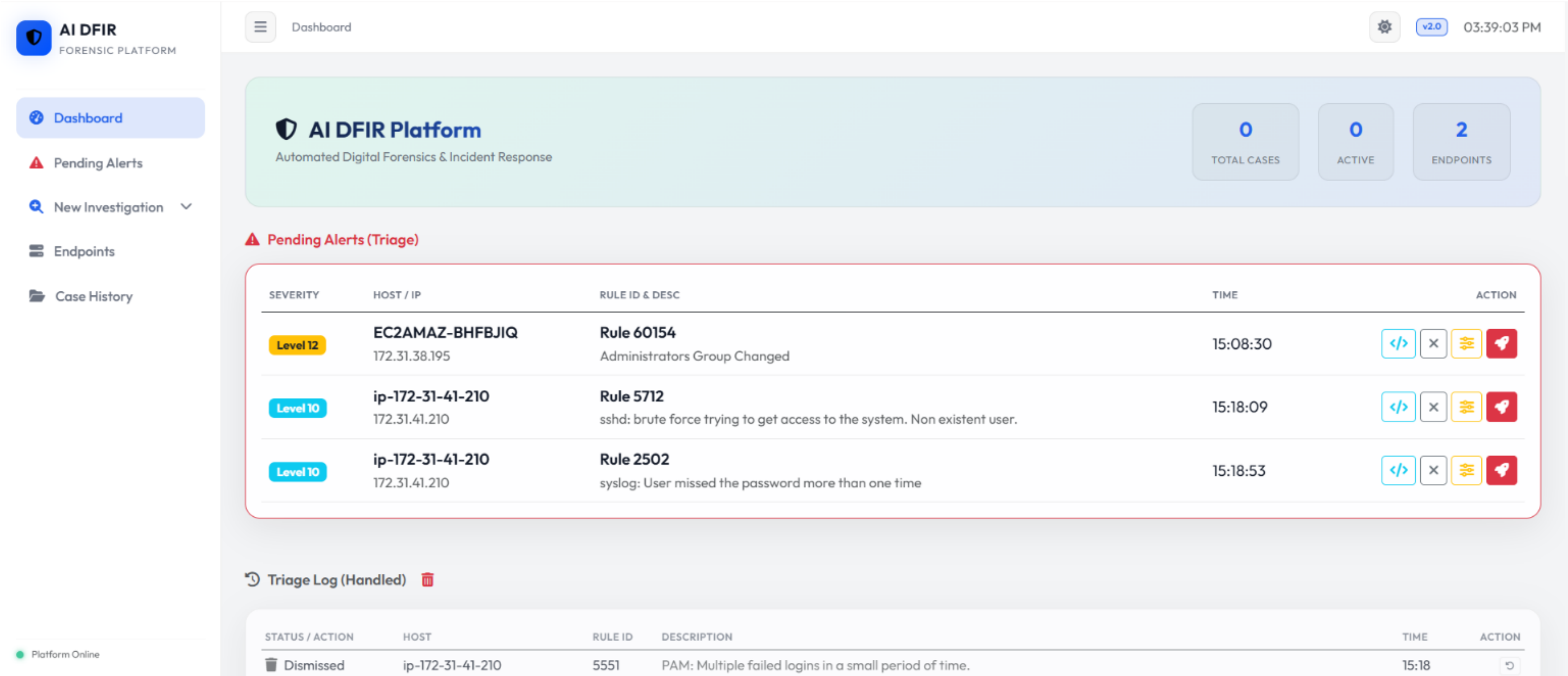


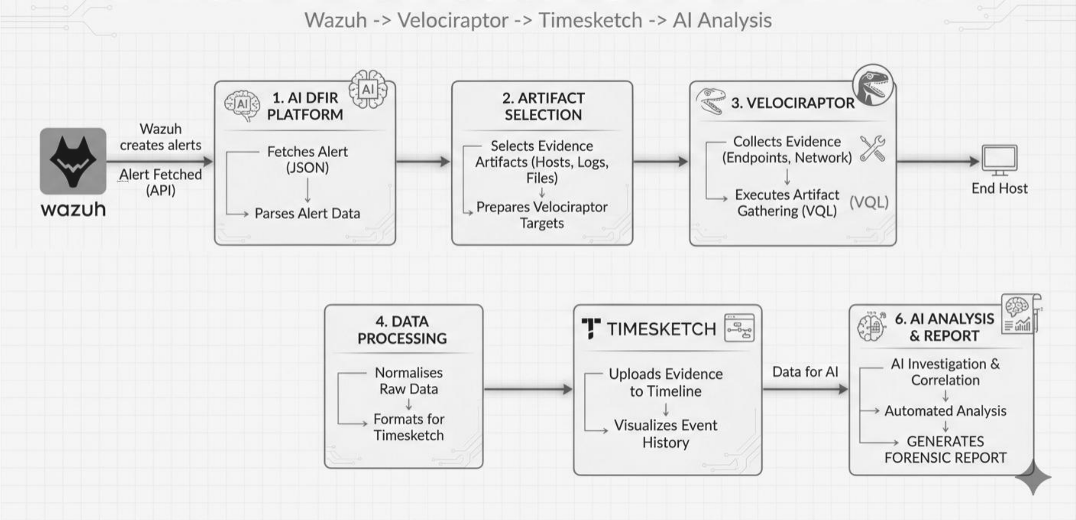
AI - Assisted Forensics

What is AI-Assisted Forensics?

AI-Assisted Forensics leverages artificial intelligence to help investigators analyze forensic artifacts, identify suspicious activity, correlate evidence, and generate actionable insights more efficiently.

Benefits

- Faster triage and investigation
- Automated artifact interpretation
- Context-aware recommendations
- Reduced analyst workload
- Improved incident response efficiency



Dashboard Tour

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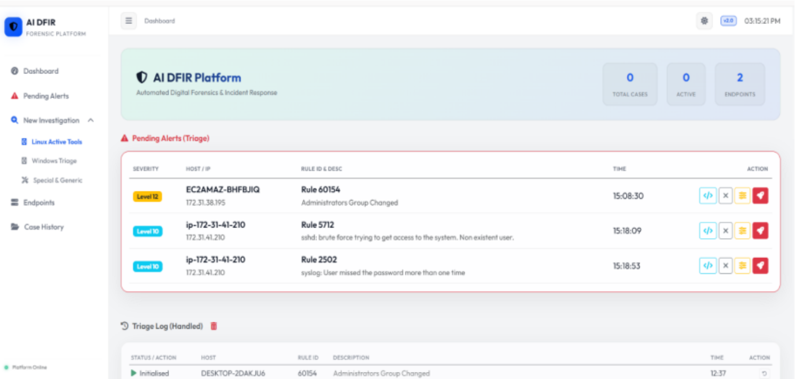

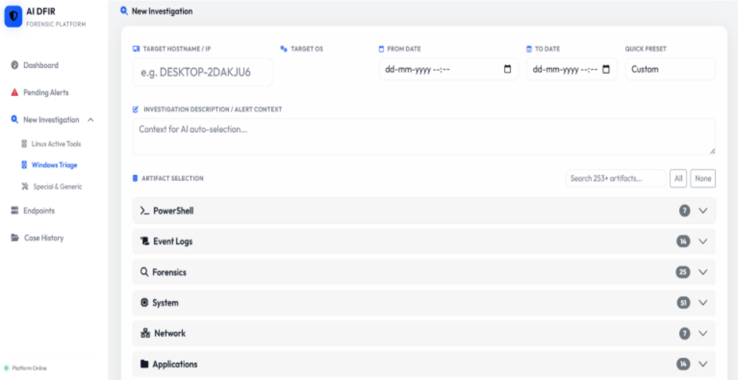

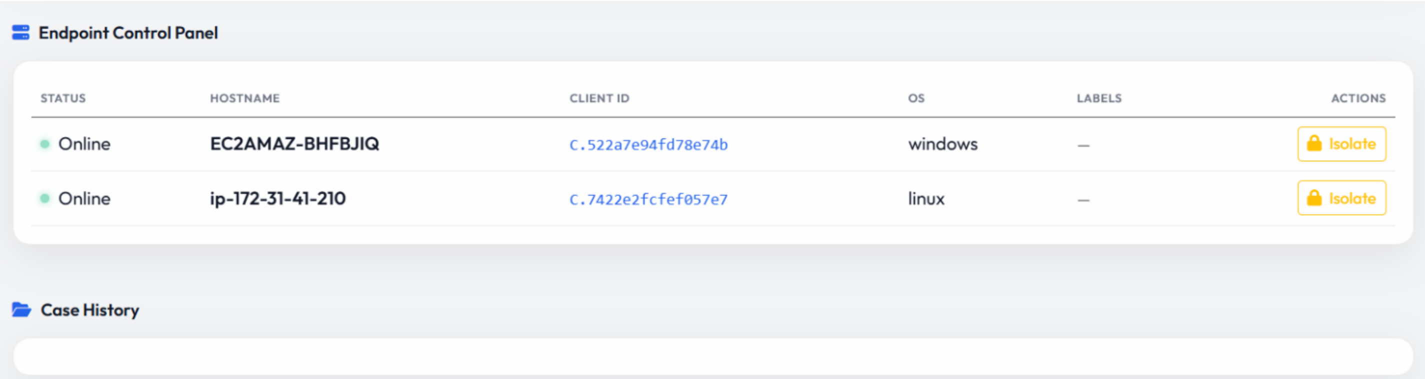

Stages of Investigation

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1

Pending Triage & Analyst Review :
Wazuh fires an alert and sends it via webhook to the AI-DFIR dashboard. The analyst sees it in the Pending Alerts queue and decides whether to start the pipeline. The dashboard maps the Wazuh rule to the right Velociraptor artifacts and detects the time range for collection.

2

Host Isolation:
The compromised host is quarantined from the network immediately. This prevents the attacker from continuing activity, exfiltrating data, or destroying evidence

3

Live Collection:
Velociraptor connects to the isolated host and collects live forensic artifacts using VQL queries. This includes running processes, network connections, file activity, event logs, registry entries, and memory artifacts. The time range is passed as filters so only relevant data is collected

4

Secure Storage & Integrity Hashing:
All collected raw artifacts are stored securely. SHA256 hashes are computed for each file to ensure chain of custody.

Stages of investigation

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5

Normalization:
Raw collected data comes in different formats and schemas. This stage flattens nested JSON structures, extracts timestamps into a consistent format, and prepares everything for timeline analysis.

6

Timesketch Upload
Normalized data is uploaded to Timesketch and organized into a multi-source timeline. This gives the analyst a visual chronological view of all events across processes, network, files, and logs in one place.

7

AI Analysis The timeline and artifacts are passed to an LLM for context enrichment. The AI identifies Indicators of Compromise (IOCs), maps activity to MITRE ATT&CK tactics, highlights suspicious patterns, and generates a narrative summary of what happened on the host.

8

Report Generation
Everything is compiled into a final PDF report — timeline, IOCs, AI analysis, artifacts, and investigator notes. This is the deliverable that goes to the SOC team or management.

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Thank you